INF10025 Data Management and Analytics

Tutorial 10 – Discussion Questions for Week 10

1. You have the following data in a relational table called customer. This data needs to be transferred to a JSON[i] document. Do this by only creating simple properties where property has a key name and a value. Do not include any property objects or arrays. Begin by only storing the data located in the first record of this file. (i.e. Zeb's information)

CustID	FirstName	Surname	DOB
100	Zeb	Kilgrave	18/04/1971
102	Trish	Walker	11/07/1984

2. Modify the JSON data that you have created so that you include a property name called **fullName**. fullName is an property object that contains **firstName** and **surname**.

3. Each customer may have **one** contact point such as mobile phone no or email address e.g. Zeb's contact details are: Type: Email Address, Details: okZeb479@gmail.com
Add another property object to the JSON document. Call it **contactInfo**.
It will contain properties **contactType** and **contactDetails**

4. Assume that each customer may have **many** contact points such as mobile phone no or email address. Add Zeb's many contacts to the JSON document within an array in the property called ContactInfo.

1	100	Home Phone	92148000
2	100	Mobile Phone	0480 133 291
3	100	Email Address	ozeb479@gmail.com
4	101	Email Address	trishtalk86@gmail.com

5. We finally need to convert Trish's data into JSON format. We could create a new document for Trish. Alternatively, we could add Trish's details to the existing document. Add Trish's details to the existing document. Create a property called Customers which contains an array.

6. Whenever you search for the differences between RDBMS and Document Databases the word 'schema' seems to appear prominently.
- How does a schema differ in a RDBMS vs Document Database?
- Why as a RDBMS often get described as a Schema on Write?
- Why does a Document Database often get described as a Schema on Read?

[i] **JSON** (JavaScript Object Notation) "is a lightweight data-interchange format. It is easy for humans to read and write. It is easy for machines to parse and generate. It is based on a subset of the JavaScript Programming Language" (json.org)